WORKED SOLUTION

STEP III 2006, Question 5

Pure • Complex Numbers and Geometry

STEP III 2006, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s3-q5

The key idea

A triangle is equilateral exactly when one side, viewed from a vertex, is another side turned through $60°$. In complex numbers that turn is multiplication by $e^{\pm i\pi/3}$, and these two numbers are precisely the roots of $t^2 - t + 1 = 0$. Shifting the origin to one vertex turns the symmetric expression into exactly that quadratic. After that, the cubic part is Vieta's formulae, and the last part is the observation that $z \mapsto pw + q$ only rotates, scales and shifts the picture.

Solution

The equilateral condition

Write $u = \beta - \alpha$ and $v = \gamma - \alpha$, the two sides leaving the vertex α. Since the points are distinct, $u \neq 0$, $v \neq 0$ and $u \neq v$. First,

$$\alpha^2 + \beta^2 + \gamma^2 - \beta\gamma - \gamma\alpha - \alpha\beta = \tfrac{1}{2}[(\alpha - \beta)^2 + (\beta - \gamma)^2 + (\gamma - \alpha)^2],$$

as expanding the right-hand side shows. Now $\alpha - \beta = -u$, $\gamma - \alpha = v$ and $\beta - \gamma = u - v$, so the bracket is $u^2 + (u - v)^2 + v^2 = 2(u^2 - uv + v^2)$. Hence

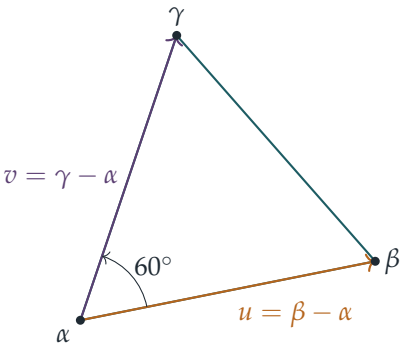
$$\alpha^2 + \beta^2 + \gamma^2 - \beta\gamma - \gamma\alpha - \alpha\beta = u^2 - uv + v^2 = u^2(t^2 - t + 1), \qquad t = \frac{v}{u}.$$

Since $u \neq 0$, the expression vanishes if and only if $t^2 - t + 1 = 0$, that is,

$$t = \frac{1 \pm i\sqrt{3}}{2} = e^{\pm i\pi/3}.$$

Ask yourself Why does $\dfrac{\gamma - \alpha}{\beta - \alpha} = e^{\pm i\pi/3}$ say exactly that the triangle is equilateral?

The modulus of v/u is $|\gamma-\alpha|/|\beta-\alpha|$ and its argument is the angle at α from side $\alpha\beta$ to side $\alpha\gamma$. So $v/u = e^{\pm i\pi/3}$ says: two sides from α are equal and the angle between them is $60°$. Such an isosceles triangle has base angles $\frac{1}{2}(180° - 60°) = 60°$, so it is equilateral. Conversely, if the triangle is equilateral then $|v| = |u|$ and the angle at α is $60°$, turned one way or the other according to the orientation, so $v/u = e^{i\pi/3}$ or $e^{-i\pi/3}$.



So the distinct points form an equilateral triangle (in either order) if and only if the expression is zero, as required.

The cubic

Let the roots of $(*)$ be α, β, γ. By Vieta's formulae, $\alpha+\beta+\gamma = -a$ and $\alpha\beta+\beta\gamma+\gamma\alpha = b$. So

$$\begin{aligned}\alpha^2+\beta^2+\gamma^2-\beta\gamma-\gamma\alpha-\alpha\beta &= (\alpha+\beta+\gamma)^2 - 3(\alpha\beta+\beta\gamma+\gamma\alpha)\\ &= a^2 - 3b.\end{aligned}$$

By the first result, the (distinct) roots form an equilateral triangle if and only if $a^2 - 3b = 0$, i.e. $a^2 = 3b$.

One caveat: the first result needs distinct points. If two roots coincide, say $\alpha = \beta$, the expression reduces to $(\gamma-\alpha)^2$, so it vanishes only when all three roots are equal. Thus $a^2 = 3b$ with repeated roots happens only for a triple root, $z^3 + az^2 + bz + c = (z + \frac{a}{3})^3$, i.e. $c = a^3/27$ as well. Apart from this degenerate "triangle" of one point, $a^2 = 3b$ is exactly the condition.

The transformation

Substitute $z = pw + q$ into $(*)$ and divide by $p^3 \neq 0$:

$$w^3 + \frac{3q+a}{p}\,w^2 + \frac{3q^2+2aq+b}{p^2}\,w + \frac{q^3+aq^2+bq+c}{p^3} = 0.$$

So

$$A = \frac{3q+a}{p}, \qquad B = \frac{3q^2+2aq+b}{p^2},$$

and therefore

$$A^2 - 3B = \frac{(9q^2 + 6aq + a^2) - (9q^2 + 6aq + 3b)}{p^2} = \frac{a^2 - 3b}{p^2}.$$

If the roots of $(*)$ form an equilateral triangle then they are distinct and $a^2 = 3b$, so $A^2 = 3B$. The roots of $(**)$ are $w = (z - q)/p$ for the roots z of $(*)$; as $p \neq 0$ this map is one-to-one, so they are distinct too. By the previous result, the roots of $(**)$ form an equilateral triangle.

Geometrically, this is no surprise: $w = (z - q)/p$ translates by $-q$, rotates by $-\arg p$ and scales by $1/|p|$, and none of these spoils an equilateral triangle. In the language of the first part, each side is divided by p, so the ratio v/u is unchanged.

Checks

- The cube roots of unity $1, \omega, \omega^2$ form an equilateral triangle and are the roots of $z^3 - 1 = 0$, with $a = b = 0$, so $a^2 = 3b$ holds.
- With $\alpha = 0$, $\beta = 1$ and $\gamma = \frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i}$, the expression is $1 + \gamma^2 - \gamma$, which is zero because $\gamma = \mathrm{e}^{\mathrm{i}\pi/3}$ is a root of $t^2 - t + 1$.
- A numerical check over 1000 random triples confirmed $\sum \alpha^2 - \sum \alpha\beta = a^2 - 3b$ for the cubic with those roots. The same check confirmed $A^2 - 3B = (a^2 - 3b)/p^2$ and that equilateral roots stay equilateral under random p and q.

Common slips

- Only checking one orientation: $\mathrm{e}^{\mathrm{i}\pi/3}$ and $\mathrm{e}^{-\mathrm{i}\pi/3}$ both appear, and the "if and only if" needs both.
- Forgetting that $u \neq 0$ (distinct points) is what lets us divide by u^2.
- Getting B wrong in the substitution: the w coefficient collects $3q^2$ from the cube, $2aq$ from the square and b, all over p^2.

Beyond STEP

The maps $z \mapsto pz + q$ are the direct similarities of the plane. Quantities they leave unchanged, like the ratio $(\gamma - \alpha)/(\beta - \alpha)$ or the vanishing of $a^2 - 3b$, are *invariants*, a central idea in geometry and algebra. The expression $a^2 - 3b$ is (up to a factor) the discriminant of f', where $f(z) = z^3 + az^2 + bz + c$. This links to a pretty fact, Marden's theorem: the roots of f' are the foci of the ellipse that touches the sides of the triangle of roots of f at their midpoints. For an equilateral triangle that ellipse is a circle, and its two foci merge into the double root of f'. Needham's *Visual Complex Analysis* develops complex geometry in exactly this spirit.

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